Foundation of Responsible AI*

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***Abstract*—**

Machine learning systems trained on historically biased data can reproduce and amplify discrimination against protected groups. Many bias mitigation methods have been proposed, but their behavior is often judged on a single dataset with an unknown amount of bias, which makes it difficult to separate the effect of the method from the effect of the data. This paper presents a controlled comparison of four representative mitigation strategies for binary classification: reweighing (pre-processing), a demographic-parity-regularized logistic regression (in-processing), and group-specific threshold adjustment for demographic parity and for equal opportunity (post-processing). Experiments use a synthetic dataset in which label bias and proxy features are explicitly parameterized, and results are averaged over 30 random seeds. At moderate injected bias, the unmitigated baseline has a demographic parity difference of 0.130 and a disparate impact ratio of 0.610. Post-processing for demographic parity reduces the difference to 0.014 at a cost of 3.4 accuracy points, whereas reweighing yields only a small improvement (0.110) with almost no accuracy loss. Crucially, methods that target demographic parity increase the equalized-odds gap (from 0.141 to 0.312 for post-processing), and the equal-opportunity method does the reverse, so no single method improves all fairness criteria. The study quantifies this incompatibility and offers practical guidance on matching a mitigation method to the fairness criterion that an application actually requires.

***Keywords*—** algorithmic fairness, bias mitigation, responsible AI, foundation of responsible AI, demographic parity, equalized odds, fairness–accuracy trade-off

1 Introduction

Automated decision systems are now used in credit scoring, hiring, criminal justice, and healthcare. When such systems are trained on records that reflect past discrimination, they can learn and perpetuate it, even if the protected attribute is not an input [1, 2]. This is commonly caused by *proxy features*, which are correlated with group membership, and by *label bias*, where historical outcomes are themselves the product of unequal treatment.

A large literature offers mitigation techniques that act before training (modifying data), during training (modifying the objective), or after training (modifying predictions) [1]. Open-source toolkits such as AI Fairness 360 [3] and Fairlearn [4] make these methods easy to apply. However, practitioners face two difficulties. First, there are several competing formal definitions of fairness, and it is known that several of them cannot be satisfied simultaneously except in degenerate cases [5, 6]. Second, empirical comparisons on real datasets confound the method with unknown properties of the data, such as how much of the observed disparity is due to label bias.

This paper addresses the second difficulty with a controlled study. We generate data in which the strength of label bias and the presence of proxy features are known, then compare four mitigation approaches along accuracy and three group-fairness metrics. The contributions are:

1. A reproducible synthetic benchmark with explicit control over label-bias strength and proxy leakage.
2. A side-by-side evaluation of pre-, in-, and post-processing mitigation across demographic parity, equalized odds, and disparate impact.
3. An empirical demonstration of the incompatibility between demographic parity and equalized odds under biased labels, and an analysis of how each method's behavior changes as bias grows.

2 Related Work

2.1 Fairness definitions

Group fairness criteria compare predictive behavior across protected groups. *Demographic parity* requires equal positive-prediction rates across groups, and the related *disparate impact* ratio underlies the "four-fifths" rule of thumb [7]. *Equalized odds* and *equal opportunity* require equal error rates, or equal true-positive rates, across groups [8]. Individual fairness instead requires that similar individuals receive similar outcomes [9]. Kleinberg et al. [5] and Chouldechova [6] showed that calibration and error-rate balance cannot in general hold together when base rates differ, and Corbett-Davies and Goel [10] critically review the limits of these metrics.

2.2 Mitigation methods

Pre-processing methods change the training data. Reweighing assigns instance weights so that the protected attribute and label become statistically independent in the weighted

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data [11], and other approaches repair feature distributions [7]. *In-processing* methods add a fairness term or constraint to the learning objective [12]. *Post-processing* methods adjust a trained model's outputs, for example by choosing group-specific decision thresholds to satisfy equal opportunity [8]. Surveys of sources of bias and the corresponding remedies are given in [1] and the textbook [13].

3 Methodology

3.1 Problem setting

Let $A \in \{0, 1\}$ denote a binary protected attribute (A=1 is the disadvantaged group), $X \in \mathrm{R}^5$ the features, and $Y \in \{0, 1\}$ the label. A classifier outputs $\hat{Y}$. We assume the protected attribute is *not* used as an input feature ("fairness through unawareness"), which is common in practice and allows proxy effects to be studied.

3.2 Synthetic data generation

For each sample, $A \sim \text{Bernoulli}(0.4)$. Features $x_1, x_2, x_5 \sim \mathrm{N}(0, 1)$ are independent of A. Two proxy features are shifted by group: $x_3 \sim \mathrm{N}(-0.8A, 1)$ and $x_4 \sim \mathrm{N}(-0.5A, 1)$. A latent merit score is

$$s = 1.0\,x_1 + 0.8\,x_2 + 0.5\,x_3 + \epsilon, \quad \epsilon \sim \mathrm{N}(0, 0.7^2), \tag{1}$$

and the observed label is

$$Y = \mathbb{1}[\,s - \beta A > 0.3\,]. \tag{2}$$

The parameter $\beta \geq 0$ is the *injected label-bias strength*: for β>0, members of group A=1 are less likely to receive a positive label for the same merit. Feature x_3 contributes to merit, so even at β=0 a group difference in positive rates arises from the proxy feature; this baseline disparity is visible in Fig. 2. Each dataset has N =20,000 samples.

3.3 Mitigation methods

All methods use logistic regression as the base learner (scikit-learn [14]) on standardized features.

- **Baseline:** unmitigated logistic regression, threshold 0.5.
- **Reweighing (pre-processing):** each training instance with $(A=a, Y=y)$ receives weight $w = \frac{P(A=a)P(Y=y)}{P(A=a, Y=y)}$ [11].
- **Fair-LR (in-processing):** minimize

$$\mathrm{L}(\mathbf{w}) = \mathrm{LogLoss}(\mathbf{w}) + \lambda\,(p^{-}_{A=1} - p^{-}_{A=0})^2 + \gamma\|\mathbf{w}\|^2, \tag{3}$$

  where $p^{-}_{A=g}$ is the mean predicted probability in group g, γ=10^{-4}, and the optimization uses L-BFGS-B. The penalty weight λ controls the trade-off.
- **Post-processing (DP):** using a validation split, choose a threshold per group such that each group's selection rate matches the overall selection rate of the baseline at 0.5.

Table 1: Test performance at $\beta = 1.0$ (mean ± std over 30 seeds). Lower DPD and EOD are better; higher DI is better.

Method	Acc.	DPD	EOD	DI
Baseline	.858±.004	.130±.007	.141±.019	.610±.017
Reweighing	.856±.004	.110±.008	.176±.018	.659±.019
Fair-LR (λ=5)	.846±.005	.078±.008	.227±.017	.738±.024
Post-proc (DP)	.824±.008	.014±.009	.312±.021	.950±.032
Post-proc (EOpp)	.870±.004	.203±.017	.036±.026	.421±.041

- **Post-processing (EOpp):** choose a threshold per group such that each group's true-positive rate matches the overall baseline true-positive rate [8].

3.4 Metrics and protocol

Let SR_g, TPR_g, FPR_g be the selection rate, true-positive rate, and false-positive rate in group g. We report accuracy; the demographic parity difference DPD = $|SR_1 - SR_0|$; the equalized-odds gap EOD = $\max(|TPR_1 - TPR_0|, |FPR_1 - FPR_0|)$; and the disparate impact ratio DI = $\min(SR_0, SR_1)/\max(SR_0, SR_1)$. Data are split 52.5%/17.5%/30% into train/validation/test (stratified by group and label); validation data are used only for threshold selection. The main experiment uses β=1.0 and 30 random seeds; the robustness experiment varies $\beta \in \{0, 0.5, 1, 1.5, 2\}$ with 20 seeds each. Results are mean ± standard deviation over seeds.

4 Results and Discussion

4.1 Main comparison

Table 1 reports results at β=1.0. The baseline model, despite not seeing A, has a demographic parity difference of 0.130 and a DI ratio of 0.610, well below the 0.8 level commonly used as a screening threshold, which shows that proxy features and biased labels propagate disparity.

Reweighing changes the baseline very little: DPD falls from 0.130 to 0.110 with accuracy essentially unchanged. A linear model that is already free to use the proxy features can recover most of the original predictive signal even after the weights decouple A and Y in the training distribution. *Fair-LR* with λ=5 lowers DPD to 0.078 for an accuracy loss of 1.2 points. *Post-processing for DP* nearly eliminates the disparity (DPD 0.014, DI 0.950) but is the most costly in accuracy (0.824, a 3.4-point drop from the baseline).

4.2 Fairness criteria conflict

The most important pattern in Table 1 is that the methods that reduce DPD *increase* EOD: from 0.141 (baseline) to 0.176, 0.227, and 0.312 for reweighing, Fair-LR, and post-processing (DP). Conversely, post-processing for equal opportunity drives EOD down to 0.036 but raises DPD to 0.203

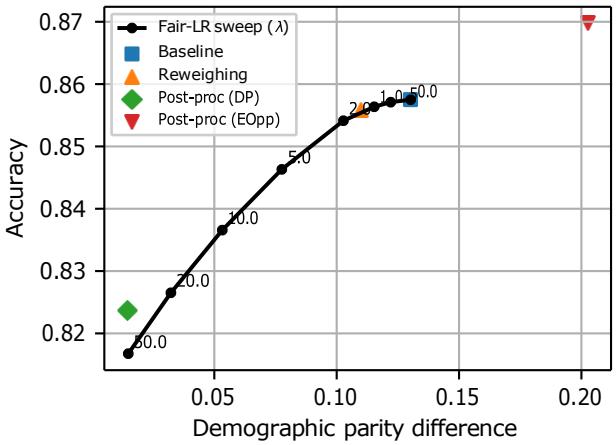


Figure 1: Accuracy versus demographic parity difference at β=1.0. The black curve is the Fair-LR sweep over λ (labels show λ); markers show the other methods (mean of 30 seeds).

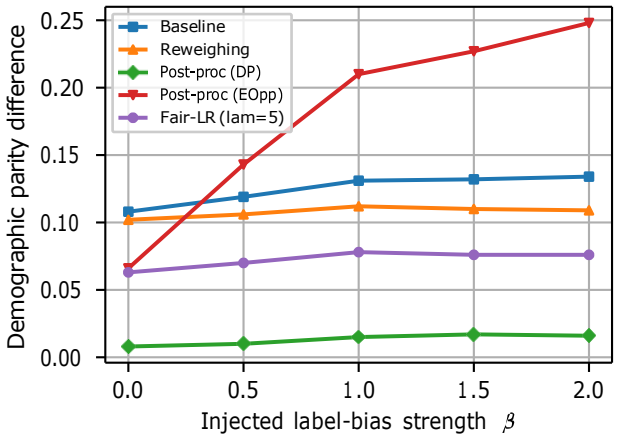


Figure 2: Demographic parity difference as the injected label-bias strength β increases (mean of 20 seeds per point).

and lowers DI to 0.421. This is consistent with theoretical results that parity in selection rates and parity in error rates cannot both hold when the groups' base rates differ [5,6]: forcing equal selection rates in a setting with unequal true prevalence necessarily distorts error rates. Choosing a "fairest" method therefore requires first choosing the criterion that matches the application, which is a normative and legal decision and not solely a technical one.

4.3 Trade-off curve for in-processing

Fig. 1 shows the Fair-LR sweep over $\lambda \in \{0, 0.5, 1, 2, 5, 10, 20, 50\}$ together with the other methods. Accuracy decreases smoothly as DPD decreases, from 0.858 at λ=0 to 0.817 at λ=50, where DPD is 0.015. At comparable DPD, Fair-LR (λ=50: DPD 0.015, accuracy 0.817) and post-processing (DP) (DPD 0.014, accuracy 0.824) lie close to each other; the post-processing point has slightly higher accuracy in our runs, with the difference being small relative to seed-to-seed variability for the in-processing sweep (std $\approx$ 0.005). The sweep also gives practitioners a tunable operating point, which a single post-processing rule does not.

4.4 Sensitivity to label-bias strength

Fig. 2 plots DPD against β. The baseline DPD rises from 0.108 at β=0 (disparity from the proxy feature alone) to about 0.13 at $\beta \geq 1$ and then plateaus; we did not isolate the reason for the plateau. Post-processing (DP) holds DPD between 0.008 and 0.017 at every β, while accuracy measured against the biased labels falls from 0.852 to 0.793 as β grows; the accuracy loss reflects the widening gap between the (biased) labels and a parity-constrained decision rule. Post-processing (EOpp) behaves in the opposite direction: its DPD grows from 0.066 to 0.248. Reweighing and Fair-LR remain roughly flat across β (0.102–0.110 and 0.063–0.078, respectively).

4.5 Practical guidance

The results suggest the following. (i) If the goal is to equalize selection rates, e.g. for outreach or screening where the disparate-impact ratio matters, a tunable in-processing penalty or a group-threshold rule is effective, but the practitioner should expect higher error-rate disparity. (ii) If the goal is equal error rates, as in settings where false negatives are costly for the disadvantaged group, equal-opportunity post-processing is effective, but selection-rate disparity will remain or increase. (iii) Lightweight pre-processing alone, such as reweighing, should not be relied upon when strong proxy features are available. (iv) Fairness should be audited on several metrics at once, since improving one can silently worsen another.

4.6 Limitations

This study has several limitations. The data are synthetic and the bias mechanism is a simple additive shift, so absolute numbers will differ on real data; the contribution is the controlled comparison of relative behavior, not a claim about any particular real-world system. Accuracy is measured against labels that are themselves biased, so a higher accuracy under bias is not evidence of a more just decision. Only one binary protected attribute, one linear base learner, and group-fairness metrics were examined; intersectional groups, nonlinear models, and individual fairness remain for future work. Confidence intervals were not computed beyond the reported standard deviations.

5 Conclusion

This paper compared pre-, in-, and post-processing bias mitigation under a controlled synthetic benchmark with explicit label bias and proxy features. Every method that improved demographic parity worsened equalized odds, and the equal-opportunity method did the opposite, confirming empirically that fairness criteria trade off against one another and against accuracy. We recommend that practitioners select the fairness criterion first, report several metrics, and treat mitigation as a tunable trade-off. Future work will replicate the study on public tabular benchmarks, extend it to intersectional groups and non-linear models, and examine mitigation under distribution shift.

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